S. HARI HARA SUDHAN

B.Tech – Chemical Engineering
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CAREER OBJECTIVE

Driven final-year Chemical Engineering undergraduate with Honours in Petroleum Process Technology and hands-on industrial internship experience spanning manufacturing, biofuel production, and cement plant operations. Demonstrated ability to identify real-world engineering problems and deliver prototype solutions. Seeking a process engineering or engineer trainee role in the petrochemical and fertilizer industry, applying strong fundamentals in petrochemical processes and plant safety to contribute to efficient, reliable, and sustainable chemical production.

EDUCATIONAL QUALIFICATION

Bachelor of Technology (B.Tech) in Chemical Engineering – Honours: Petroleum Process Technology

2023 – Present
Erode Sengunthar Engineering College (Anna University), Erode, India — CGPA: 8.1 / 10.0

- Relevant Coursework: Thermodynamics, Fluid Mechanics, Mass & Heat Transfer, Process Calculations, Chemical Reaction Engineering, Petroleum Refining Technology, Petrochemical Processes

Higher Secondary Certificate (HSC)

2022 – 2023
Nanjappa Municipal Corporation Boys Higher Secondary School, Tiruppur, India — Score: 67.5%

TECHNICAL SKILLS

- Process Simulation: MATLAB, Aspen Plus (Beginner), DWSIM (Beginner)
- Office & Reporting: MS Excel, MS PowerPoint
- Laboratory: Titration, Distillation, Fermentation, pH Analysis, Proximate and Ultimate Analysis
- Plant Knowledge: Cement Manufacturing, Bioethanol Pilot Plant, Cable & Wire Manufacturing
- Certification: NPTEL – Waste to Energy Conversion (Elite Grade)

PROJECTS

Wire Diameter Analyser – Industrial Prototype (funded ₹6000)

6th Semester
Goflex Wires and Cables (CRI Pumps), Coimbatore

- Designed and built a prototype to automate wire diameter measurement and instantly flag deviations from standard specifications, addressing a live production-line challenge.
- Reduced reliance on manual inspection, improving quality assurance accuracy and minimising material waste.

Design and Simulation of Thermal Insulation in Multi-Layer Composite Walls for Energy-Efficient Buildings

5th Semester
Erode Sengunthar Engineering College

- Designed and simulated multi-layer composite wall configurations to evaluate thermal insulation performance for energy-efficient building design.

Bioethanol Production from Biomass Waste – Mini Research Project

4th Semester
Sathyabama University, Chennai

- Studied sustainable bioethanol production pathways including feedstock preparation, fermentation optimisation, and ethanol recovery techniques.
- Delivered team project documentation and technical presentation, reinforcing collaboration and communication skills.

Valorization of Silica-Rich Agricultural Waste into Low-Cost Adsorbents for Wastewater Treatment *Ongoing*

- Researching biogenic silica derived from rice husk ash and sugarcane bagasse ash as low-cost adsorbents for wastewater treatment.

INTERNSHIP EXPERIENCE

Manufacturing Process & Quality Engineering Intern (Stipend ₹15000) *t*

Jan 2026 – Mar 2026

Goflex Wires and Cables (A Unit of CRI Pumps), Coimbatore, India

- Learned raw material handling, cable extrusion, and quality inspection on the production floor.
- Spotted wire diameter inconsistencies in real time and built an automated prototype to catch production variations.
- Followed plant safety guidelines and standard operating procedures (SOP) at all times.

Bioethanol Pilot Plant Intern

Feb 2025 – Apr 2025
International Research Center, Sathyabama University, Chennai, India

- Monitored fermentation and distillation in real time and helped run equipment parameter tests.
- Carried out routine sample analysis, including titration, pH testing, and distillation checks.

Process Overview & Plant Operations Intern (15 Days)

Oct 2024
India Cement Pvt. Ltd. (Sankari Plant), Salem, India

- Observed large-scale cement manufacturing, from raw material preparation to final clinker cooling.
- Learned the basics of kiln operations, grinding mills, and emission control.

BOOK PUBLICATION

"Enhancing Bio-Based Separation Processes: The Potential of Pass-Through Distillation"

20 Apr 2025

- Article published as part of the Sathyabama International Conference proceedings.

CO-CURRICULAR ACTIVITIES

- Member – Indian Institute of Chemical Engineers (IIChE), Student Chapter
- Presented a model on "Green Fuels" at University Tech Fest, demonstrating research communication and public speaking skills.
- Volunteer – Clean Energy Awareness Drives, promoting sustainability and safe energy practices.

ACHIEVEMENTS

- 3rd Place – Paper Presentation, International Conference, Sathyabama University, Chennai
- 2nd Place – Paper Presentation, Mahindra Engineering College
- Participated in workshops on Process Safety, Arduino, SAP, and Engineering Design
- NPTEL – Waste to Energy Conversion (Elite Grade) certification

PERSONAL DETAILS

- Nationality: Indian
- Date of Birth: 19.03.2005
- Languages Known: Tamil (Professional), English (Professional), Hindi (Beginner)

DECLARATION

I hereby declare that the information furnished above is true to the best of my knowledge and belief.

Date: _____
Place: Tirupur, Tamil Nadu

S. Hari Hara Sudhan